

2 (**short**) A fire extinguisher contains a perfect gas with $c_p = 1000 \text{ J kg}^{-1} \text{ K}^{-1}$ and $\gamma = 1.25$ at 300 K and 10 bar. The gas is expanded adiabatically through a valve to 1 bar.

(a) If the process were isentropic, what would the final temperature of the gas be? [2]

(b) The process is not isentropic. Sketch the process on a T - s diagram, labelling all lines clearly. [3]

(c) If the final temperature of the gas is 246 K, calculate the entropy change per unit mass of gas and the ratio of the final volume of the gas to its initial volume. [5]